subject to the orthogonality constraint $\boldsymbol{\Theta}^T\boldsymbol{\Theta} = \boldsymbol{I}$. Here the potential energy has been expressed in terms of $\boldsymbol{q}_{\mathrm{cm}}$ and the orientation of the body. The constraint may be enforced using Lagrange multipliers with the following equations

$$\begin{aligned} \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{q}_{\mathrm{cm}} &= \boldsymbol{p}_{\mathrm{cm}}/M, \quad \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Theta} = \boldsymbol{\Pi}\boldsymbol{R}^{-1}, \\ \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{p}_{\mathrm{cm}} &= -\nabla_{\boldsymbol{q}_{\mathrm{cm}}} U, \; \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Pi} = -\nabla_{\boldsymbol{\Theta}} U - \boldsymbol{\Theta}\boldsymbol{\Lambda}, \\ & \boldsymbol{I} = \boldsymbol{\Theta}^T\boldsymbol{\Theta}. \end{aligned}$$

These equations may be solved using a SHAKE or RATTLE algorithm [84, 201]. However, we will present an alternative which is both conceptually simpler and easier to implement.

4.4.1 Euler Equations

An elegant form of the equations of motion for the free rigid body is expressed in terms of the angular momentum vector referred to the body frame.

The angular momentum vector in the reference frame is given by

$$\boldsymbol{l} = \sum_{i=1}^{\mathrm{N}} m_i \boldsymbol{\delta}_i \times \dot{\boldsymbol{\delta}}_i.$$

From Eq. (4.39) we may write this as

$$\boldsymbol{l} = \sum_{i=1}^{\mathrm{N}} m_i \boldsymbol{\delta}_i \times \boldsymbol{\omega} \times \boldsymbol{\delta}_i,$$

from which, using the vector triple product identity $\boldsymbol{a} \times \boldsymbol{b} \times \boldsymbol{c} = (\boldsymbol{a} \cdot \boldsymbol{c})\boldsymbol{b} - (\boldsymbol{a} \cdot \boldsymbol{b})\boldsymbol{c}$, we obtain

$$\boldsymbol{l} = \sum_{i=1}^{\mathrm{N}} m_i \left[\|\boldsymbol{\delta}_i\|^2 \boldsymbol{I} - \boldsymbol{\delta}_i \boldsymbol{\delta}_i^T \right] \boldsymbol{\omega}.$$

Define $\boldsymbol{\pi} = \boldsymbol{\Theta}^T\boldsymbol{l}$, the angular momentum vector referred to the body frame of reference, then we have (using $\boldsymbol{\delta}_i = \boldsymbol{\Theta}\delta_i(0)$):

$$\boldsymbol{\pi} = \boldsymbol{T}\boldsymbol{\omega}_{\mathrm{b}}.$$

Thus the inertia tensor relates the angular momentum vector in the body frame to the angular velocity vector in the body frame.